

Substation Name: Klipfontein 132/33 kV Substation

Location: Farm Klipfontein 24 IR, near Benoni, Gauteng Province

Coordinates: 26°11'24" S, 28°18'36" E

Purpose: Supply new residential + light-industrial growth in the Greater Johannesburg East area and offload the overloaded Rynfield and Daveyton 132/33 kV substations.

Client: Ekurhuleni Metropolitan Municipality (as distributor) with Eskom Transmission as bulk supplier.

Compliance Framework: All documents reference the South African Grid Code, NRS 048, SANS 1019, IEC 60076 / 62271, IEEE 80 (earthing), and Eskom-specific TRS/STD specifications. Below is the supplied folder structure and content you will copy into separate PDF documents.1.

Project Documents

/Data Provided/

- 01_Network_Planning_Report.pdf
- 02_Equipment_&_Primary_Plant_Specification.pdf
- 03_Geotechnical_Report.pdf
- 04_Environmental_Impact_Assessment_Report.pdf
- 05_System_Studies_Report.pdf
- 06_Line_Design_Report.pdf
- 07_Grid_Connection_Requirements.pdf
- 08_Project_Budget_&_Scope.pdf
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01_Network_Planning_Report.pdf

This report is produced by Eskom Transmission System Planning / Distribution Planning Department after load-forecast modelling and network studies. It defines exactly where the new substation fits into the Gauteng East Grid and what performance is mandatory. It is the document that drives every later decision on fault levels, redundancy, and feeder numbers. You will refer to it constantly during the Design Basis Report.

Key Data

Short-Circuit Fault Levels (3-phase and 1-phase, calculated per IEC 60909 and SANS 1019):

Voltage	3-Phase Fault (kA)	1-Phase Fault (kA)	X/R Ratio
132 kV	31.5	25	15
33 kV	20	16	12

Load Forecast (MVA at 33 kV bus, 0.9 pf):

Year 1 (2027): 65 MVA

Year 5 (2031): 98 MVA

Year 10 (2036): 135 MVA

Feeder Requirements:

2 × 132 kV incoming lines (from different sources for N-1).

Initial 8 × 33 kV outgoing feeders; provision for 12 ultimate.

Reliability & Security:

Full N-1 compliance (no load loss for single circuit or transformer outage). Urban category – medium-high security per Grid Code.

Future Extension:

Space reserved for +4 feeders and possible 3rd transformer.

02_Equipment_&_Primary_Plant_Specification.pdf

Issued jointly by Eskom Plant Engineering and the selected OEM (e.g., ABB / Siemens / CG). It gives the exact ratings the equipment must meet to survive the network conditions above. All ratings are selected for 40 °C ambient, 1 000 m altitude, and lightning level of 1.2/50 µs wave.

Transformers

Quantity: 2 × 75 MVA ONAN/ONAF

Voltage: 132/33 kV, vector YNd11

Impedance: 12 % ± 7.5 %

OLTC: ±10 % (17 steps), 1 500 A rated

Tertiary winding: 11 kV, 5 MVA (for possible neutral reactor or auxiliary supply)

Insulation level (BIL): 550 kV (132 kV) / 170 kV (33 kV) / 95 kV (tertiary) per SANS 1019 & IEC 60076.

Busbar Ratings

132 kV: 3 150 A continuous, 31.5 kA / 3s short-time

33 kV: 2 500 A continuous, 20 kA / 3s

Shunt Reactors & Capacitors

132 kV line shunt reactor: 20 MVar (to control Ferranti effect on long lines)

Neutral reactor: 400 Ω (for 132 kV reactor neutral)

33 kV capacitor bank: 10 MVar (switched, for voltage support)

03_Geotechnical_Report.pdf

Prepared by a registered geotechnical engineering firm (e.g., Jones & Wagener or similar) after site investigation. The electrical engineer does not do the Wenner test — the geo team does it using ASTM G57 / SANS equivalent before you start earthing design. This report is critical for foundation sizing, earth-grid design (per SANS 10199 & IEEE 80), and civil works.

Soil Resistivity (Wenner 4-electrode method, average wet-season values):

Depth (m)	Resistivity ($\Omega\cdot\text{m}$)
0.5	115
1.0	92
2.0	78
3.0	65

Average design value:

85 $\Omega\cdot\text{m}$ (seasonal correction factor 1.2 applied).

Soil type: Clayey sand with high moisture retention (good for earthing).

Bearing capacity: 180 kPa (safe for transformer foundations).

Groundwater: 2.8 m below surface in wet season.

Recommendation:

Use deep-driven rods + counterpoise where resistivity > 100 $\Omega\cdot\text{m}$ in dry pockets.

04_Environmental_Impact_Assessment_(EIA)_Report.pdf

Compiled by an independent environmental consultant under the National Environmental Management Act (NEMA) 1998 and EIA Regulations 2014. Eskom and the municipality require full Basic Assessment or Full EIA before any construction. This is a legal prerequisite — no shovels in the ground without the Record of Decision (RoD).

Key Findings

Land use: Semi-urban agricultural/smallholding – no protected vegetation or Red Data species impacted.

Noise & EMF: Within SANS 10103 and ICNIRP limits with standard mitigation.

Heritage: No sites of significance within 500 m.

Constraints: Construction hours 07:00–17:00 weekdays; dust suppression mandatory; wetland buffer 50 m on eastern boundary.

Approved mitigation: Indigenous screening trees around perimeter; bird flight diverters on incoming lines.

05_System_Studies_Report.pdf

Issued by Eskom System Operations or an approved specialist after PSCAD/ETAP/DigSILENT modelling. These studies confirm the substation will not cause instability or unacceptable voltage excursions.

Studies Included

Load-flow: Voltage profile 0.95–1.05 pu under N-1.

Transient stability: Stable for 120 ms fault clearance.
Switching studies: Controlled switching required for shunt reactors to limit inrush.
Harmonic study: THD < 5 % with proposed capacitor banks.
Ferro-resonance & temporary overvoltage: Within IEC 60071 limits.

06_Line_Design_Report.pdf

Prepared by Eskom Transmission Lines Department. Defines the incoming 132 kV circuits.

Conductor: ACSR Zebra (54/7 stranding, 420 mm² Al, overall Ø 28.6 mm, mass 1.62 kg/m, ultimate tensile strength 131 kN).
Configuration: Double-circuit vertical, 400 m span average.
Insulation: 550 kV BIL disc insulators, creepage 31 mm/kV.
Lightning performance: Expected flashover rate < 0.5/100 km/year.
Tower type: Guyed mast or self-supporting lattice (to be finalised).

07_Grid_Connection_Requirements.pdf

Direct extract from the South African Grid Code (Network Code) + Eskom-specific connection agreement. This is non-negotiable.

Requirements include: SCADA & telecom integration (IEC 61850 & DNP3).
Protection coordination with upstream (distance + backup OC/EF).
Synchro-check and auto-reclose blocking.
Power quality per NRS 048-2 (voltage, flicker, harmonics).
Fault ride-through and frequency response if generation is later added.

08_Project_Budget_&_Scope.pdf

Issued by the client (Ekurhuleni + Eskom funding model). This sets the financial envelope.
Approved Budget: ZAR 195 million (excl. VAT) – 2026/27 prices.
Scope: Full turnkey design, supply, construction, testing & commissioning of the substation and two incoming 132 kV lines (approx. 8 km total). Includes civils, earthing, control building, and 3-year defects liability.
Exclusions: Land acquisition (already secured) and future 3rd transformer.

09_Protection_Philosophy_Document.pdf

Eskom Protection & Automation standard philosophy (aligned with DST_34-462 and NRS 037).

Philosophy

132 kV lines: Main + backup distance (21/21N) with auto-reclose.
Transformers: Differential + restricted earth fault + Buchholz + OLTC supervision.
33 kV feeders: Directional OC/EF + sensitive earth fault.
Busbar: High-impedance or low-impedance differential (to be confirmed in DBR).
All relays: Numerical, IEC 61850 GOOSE, with full event recording and SCADA integration.

